New Devices, More Mobility

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The world has gone mobile. Just look around. Laptops are everywhere, Wi-Fi hotspots are in airports, hotels, and street corners, and over 200 million people in India own a cell phone! The mobile computing revolution began years ago with laptops built upon Intel Centrino technology and it has grown rapidly ever since. Now get ready for the next revolution as the Internet gets more personal with ubiquitous connectivity, new form factors that blur the lines between cell phones and laptops, and amazing new human interfaces.

One tenet for the future of mobility lies in completely ubiquitous connectivity. Not the islands of connectivity that we have with today's generation of Wi-Fi hotspots, but complete connectivity. Whether you're stuck in traffic in Bangalore or working in your village or farm, you will always be connected to a high-speed, high-bandwidth data and voice network.

WiMAX promises to deliver mobile, wireless broadband connectivity with 1 to 5 Mbps at ranges up to 70 km. Wi-Fi gives coverage like a cordless phone; WiMAX offers coverage like a cell phone. In pursuit of making WiMAX a true global standard, Intel is working with legislative and regulatory bodies worldwide to establish 2.3-2.5, 3.5, and 5 GHz as global frequency bands for WiMAX. The spectral efficiency and extended coverage characteristics of WiMAX make it ideal to provide significantly lower cost-per-data-bit mobile broadband connectivity in places where there is no other means of providing broadband connectivity. We are encouraging regulators to rapidly allocate large blocks of spectrum in a technology-neutral way.

As users get more connected, opportunities present themselves for new and exciting form factors. Today I have incredible functionality with my notebook PC, but it's just too large. Smartphones are wonderful communication devices, but are severely limited in software compatibility, are limited in bandwidth, and offer only compromised data services. The future will be the best of both without compromises. A new category of device is emerging, which blurs the line between laptops and cell phones: Ultra Mobile PCs (UMPCs). Intel and several of its partners announced plans for the first of these devices earlier this year. They will be roughly four to seven inches in size. Despite their size, these are complete personal computers including wireless connectivity, a full-fledged OS optimised for mobility, and a breadth of applications. Docked to an external display and a keyboard, these devices will feel just like your laptop does today.

> "Get ready for the next revolution: there will be new form factors and amazing new human interfaces"

PCs have made only modest advances in terms of their user interface. Smartphones are even worse; they have tiny thumb boards or stroke and stylus interfaces that are even more foreign for a human. Fundamentally, humans have needed to adapt to computers, not computers adapting to humans. While speech recognition has made progress, its use is quite narrow and its quality remains poor. Now, though, we are on the verge of having enough computing capability to provide multi-modal human interfaces that seamlessly combine multi-channel, high-quality voice with stereo vision on the input. On the output side, we're likewise soon to see high-definition sound, video, and graphics rendering systems. The combination will present the most dramatic step in human computer interfaces since the move from DOS to Windows: computers that can listen, see, and provide both semantic and contextual recognition and learning capabilities.

India's GDP is projected to grow by about 7 per cent this year, making it a prime target of growth for many goods and industries. Technology is expected to grow even faster, further fuelling disposable income within this populous nation. With 54 per cent of the Indian population below the age of 25 years, at a macroeconomic level the technologies I've described above will help drive both higher productivity and higher growth rates for the economy. Your ability to "Go Mobile" plays a critical role at many levels of the economy from the rural to the urban. As a nation, these technologies are critical to enabling the continued rise as a global IT innovator and exporter. In cities, these technologies will help professionals become more productive and help them conduct their business in a seamless fashion. In rural communities, they will help people get connected with technology advancements, furthering their businesses and educational opportunities. The future of mobile computing will enrich all our lives and will help further connect India to the worldwide economy. ■